

WrightHere: Supporting Children's Creative Writing with AI-Infused Interactive 3D Environment

Jaeryung Chung
Department of Industrial Design
KAIST
Daejeon, Republic of Korea
jhyun513@kaist.ac.kr

Seon Gyeom Kim
Department of Industrial Design
KAIST
Daejeon, Republic of Korea
ksg_0320@kaist.ac.kr

Tak Yeon Lee
Department of Industrial Design
KAIST
Daejeon, Republic of Korea
takyeonlee@kaist.ac.kr

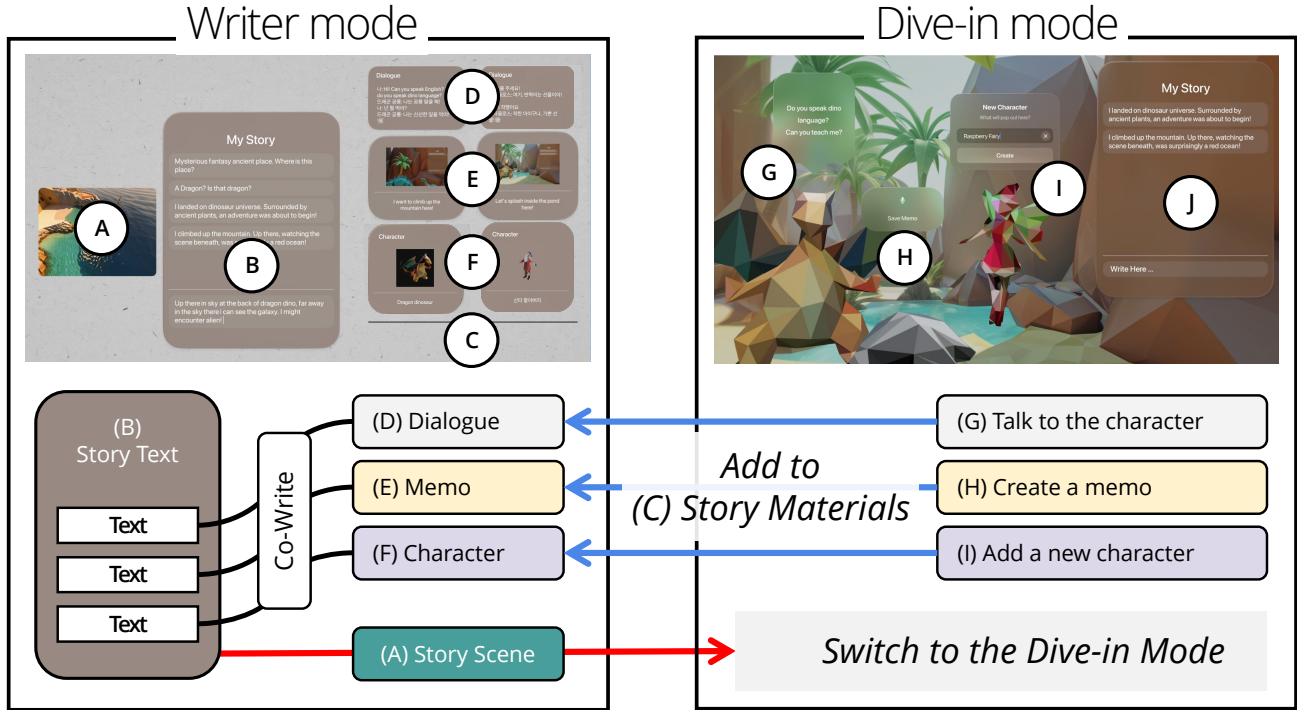


Figure 1: WrightHere's two main modes: Writer Mode and Dive-in Mode. In *Writer Mode*, users create stories in the (B) *Main Editor*, which generates corresponding 3D (A) *Story Scenes*. By clicking on the scene, users enter *Dive-in Mode* where they can interact with characters through (G) conversation, (H) create memos, or (I) add new characters. These interactions are saved as (C) *Story Materials* and displayed as (D), (E), and (F). Users can drag these *Story Materials* into the (B) *Main Editor* for co-writing suggestions. The (J) *Main Editor* remains accessible during *Dive-in Mode*, allowing users to seamlessly switch between different *Story Scenes* and continue writing.

Abstract

WrightHere is a generative AI-infused writing system that generates interactive 3D environments of the written story where users can explore, interact with characters, and gather inspiration to facilitate their creative writing. While creative writing is crucial for child

development, it poses a unique challenge and sets a high hurdle for children. Building upon past research of providing effective stimuli for new inspiration, we explore how AI-infused interactive 3D scenes of stories can spark creativity and help children maintain their writing momentum. Through user studies with the *WrightHere* system, we examined how this integration of AI-generated 3D environments with writing interfaces enhances engagement and writing output. This work presents *WrightHere* as a novel prototype exploring the potential of generative AI and interactive 3D environments in supporting children's creative writing process.

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CCS Concepts

• **Human-centered computing** → **Interactive systems and tools**.

Keywords

Generative AI, 3D interaction, creative writing, storytelling, LLM

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1 Introduction

Creative writing plays a vital role in child development [10, 11], operating through an inherently cyclical process [10, 13]. Building on Flower and Hayes’ cognitive process theory [10], this cycle moves through stages of planning, where writers form internal representations of knowledge often as abstract as fleeting images, and translation, where these ideas are converted into visible language. However, this natural flow is frequently disrupted by writer’s block [16, 18], presenting a persistent challenge in creative writing. Research has approached this challenge by providing various stimuli to break through writer’s blocks. Studies have explored guided questions [2, 8, 20], planning tools [1], visual aids through drawing [4], and group brainstorming [2, 20]. With the emergence of generative AI, these approaches have expanded to include AI-assisted writing aids [3, 16, 31], conversational support [14, 19], visual stimuli [2, 23, 28], audio elements [26], and character-centered systems facilitating story development through interaction [22, 30]. These individual modalities could enhance creative writing, however, little has been explored about their integration in multi-modal interactive 3D environments [24, 27]. Furthermore, although providing these various stimuli supported creativity, there are opportunities to better support the cyclical nature of creative writing [12, 35]. Recent language model research explored how AI can be utilized in creative writing to support this cyclical process [6, 35, 36]. Existing systems utilize selection-based interactions to support cyclical processes [9, 33], yet they do not extend to using AI-generated 3D environments.

This research gap leads us to investigate three key questions: (1) How do writers explore generative 3D environments in creative writing? (2) How can these environments be integrated into a system to support continuous cycles between imagination and composition in creative writing? (3) How can we ensure that inspiration gained from these rich environments effectively translates into written narrative, preventing ideas from remaining as abstract thoughts?

To address these questions, we present *WrightHere*, an AI-infused 3D system that reinforces the cyclical nature of creative writing. Writers can generate 3D scenes of their own narrative worlds, explore within these interactive environments, engage with characters they have created, and return to writing with newly gathered inspiration, establishing a continuous flow between imagination and story composition. The system operates through two integrated modes: *Writer Mode* for story composition and *Dive-in Mode* for

exploration. Through user studies with the *WrightHere* system, we examined how this integration of AI-generated environments with writing interfaces enhanced engagement and writing output. Participants demonstrated strong engagement with the system’s features, exploring 3D environments to develop and progress their narratives.

This work makes two primary contributions to the field of creative writing support: First, we explore the possibilities of generative AI in creating interactive 3D environments that stimulate imagination and enhance writing engagement. Second, we demonstrate how these environments can be integrated into a system that actively supports the cyclical nature of creative writing, enabling switching between imagination and story composition.

2 WrightHere System Design

WrightHere is a generative AI-infused writing system that generates interactive 3D environments of the written story where users can explore, interact with characters, and gather inspiration to facilitate their creative writing. Our system supports creative writing through two distinct but interconnected modes: *Writer Mode* and *Dive-in Mode*. This design is grounded in research [10] showing that creative writing involves two main cognitive stages: planning and translating. Users can freely switch between these modes as they develop their story. In *Writer Mode*, which supports the translating stage, users write their story in the *Main Editor*. Each paragraph automatically generates a 3D scene, shown as thumbnail images beside the text. The system provides *Story Materials* (Fig.2) of dialogues, characters, and memos that users can incorporate into their writing through a co-writing function. These materials come from their interactions in *Dive-in Mode*. *Dive-in Mode* facilitates the planning stage by letting users enter and explore the 3D environments generated from their paragraphs. Here, users can create and converse with characters, and capture ideas as memos. The system generates 3D characters based on users’ descriptions, enables dynamic conversations through ChatGPT with text-to-speech output, and saves memos along with their visual context. All these interactions are stored as *Story Materials* that become available when users return to *Writer Mode* by pressing the ESC button.

This easy switching between writing and interactive exploration allows writers to seamlessly transition between planning and translating stages of their creative process. When writers need to imagine and develop their ideas, they can enter *Dive-in Mode* to explore and interact with their story world. When they’re ready to write, they can return to *Writer Mode* where their explorations are preserved as *Story Materials*, ready to be transformed into narrative text.

The system integrates real-time AI generation for most features, with generative AI APIs creating 360-degree environments [21], text-to-speech output [7], and 3D character meshes [25]. Background music is an exception, using a preloaded library of 9 ambient tracks that are generated manually [17] to ensure consistent quality and reduce latency. ChatGPT-4o [29] serves as the central controller, analyzing scene context to select appropriate music tracks, generating prompts, managing character dialogues, and providing writing support. The entire system was developed under the Unity framework.

2.1 Writer Mode

In *Writer Mode*, users create their stories in the *Main Editor* while having a parallel view of their *Story Materials*. As users write each paragraph, the system automatically creates a 3D *Story Scene* by mapping generated 360-degree images into a 3D space and plays background music that fits the mood. When hovering onto each paragraph, users see a small preview image of the scene on the left side. Users can click on the scene to enter it in *Dive-in Mode*. When they make changes to their story, both the scene and the preview image update right away. To create these scenes, the system uses ChatGPT to analyze the background of the scene to extract key words that will go to 360-degree image generation [21]. Also for music, ChatGPT figures out the mood of each scene and picks a track that matches that feeling. The music track is selected from a preloaded library of music, where the system has nine different tracks available - three happy moods, three calm moods, and three tense moods. Music was generated through Musicfy API [17] manually and was built into the system. The *Writer Mode* saves *Story Materials* in three categories: character, dialogue, and memo. These materials are automatically generated through interactions in *Dive-in Mode*. Character materials contain both the character's written description and its visual representation. Dialogue materials store the complete conversation exchange, showing each line marked with either 'me:' or 'character:' to track who said what. Memo materials combine the user's captured idea with a snapshot of what they were looking at in the scene when they wrote it. Users can drag *Story Materials* into the *Main Editor* to use the co-writing function. The system generates one or two sentences based on the material type. For dialogue, it creates story text featuring the specified conversation. For characters, it generates a narrative centered on the character. For memos, it incorporates the memo's content into the story. The system considers the entire existing story when generating new content to maintain coherence. Users can change any of their *Story Materials* before using them. With this, they can also update character details, which helps make future conversations in 3D environments more engaging.

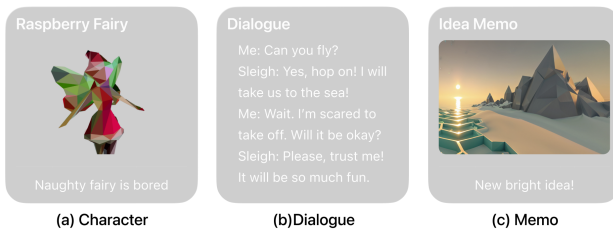


Figure 2: Story Materials

2.2 Dive-in Mode

In *Dive-in Mode*, users enter a 3D environment that represents their story paragraph, where they can explore the space with background music playing. Users can interact with the environment in three main ways. First, they can create characters by clicking anywhere in the 3D scene. This opens a UI where users describe what they want to create. They can write either sentences or keywords, which

the system processes through ChatGPT. The system converts these descriptions into detailed prompts for 3D mesh creation. The Meshy API [25] then generates the 3D character in real-time, and when it appears on scene, it shows subtle idle animations to make the character feel alive. Users can create multiple characters as much as they want in their scene. Characters remain persistent in the scene, so users can find them exactly where they left them when returning to the *Story Scene*. The second interaction happens when users hover over a character to start a conversation. Users type their messages, and ChatGPT generates responses that fit with the story, previous dialogues, and the character's described traits. The system includes any character descriptions from the creation process in its conversation prompts to maintain consistency. When a character responds to the user, it shows the text in a bubble above the character and plays generated speech through a text-to-speech API [7], making the character actually speak to the user. The third interaction is saving memo, activated by pressing the spacebar. When users capture an idea, the system saves both the memo and their current view of the scene as *Story Material*. This helps users remember the context of their ideas when they return to writing later. All these interactions are saved as *Story Materials*, but they only become visible when users return to *Writer Mode*. This design choice aims to help users stay focused in the *Dive-in Mode* experience without UI distractions. The *Main Editor* stays active, so when users write new story content, it generates a new *Story Scene* immediately. Users can enter the new scene right away if they want to continue creating and interacting.

2.3 Use Case Scenario

The use case scenario was designed to help users understand the system by incorporating all key functionalities into a cohesive narrative. A full version of it is embedded as a pre-built walk-through tutorial within the system and is used during the initial onboarding process of the preliminary user study to introduce users to the available features. In this section, a shorter version of the scenario is leveraged as a foundation to provide a clearer explanation of the system. It is presented alongside Figure 3, aligning the visuals and descriptions with the use case scenario instead of relying solely on a textual explanation, to provide a more intuitive understanding of the system. The following paragraph is the use case scenario featuring a fictional user, Sarah.

Sarah, a 10-year-old aspiring writer, is developing a story about Christmas elves on an adventure. She writes in her *Main Editor* (Fig.3.a): "They were so excited to start an adventure, so they stepped out bravely from their house." When she finds herself unsure how to continue, she enters the *Dive-in Mode*. She finds herself in a scene of a snowy outdoor environment in front of the house with gentle wind sounds in the background. As Sarah explores the space, an idea strikes her. She quickly uses the *Memo Function* (Fig.3.b) to capture her thought: "Some villain will appear soon to mess up this peace." Thinking about how her characters might travel through this winter landscape, she uses the *Character Creation Function* (Fig.3.c) to add a sleigh to the scene. "Can you fly?" she asks the sleigh through the *Dialogue function* (Fig.3.d). When it responds "Yes, hop on!" Sarah excitedly writes the next story beat: "We should take off!" After exploring several different *Story*



Figure 3: An example use case scenario of creating Christmas adventure story

Scenes (Fig.3.e) and collecting various story elements, Sarah returns to *Writer Mode*. Here, she finds all her accumulated *Story Materials* (Fig.3.f) - her memos, created characters, and dialogue exchanges. Using these materials and the *Co-writing Function*, she continues developing her adventure story with renewed creativity. When she faces another creative block while writing, she dives back into her story world, creating a cycle of exploration and writing that helps her continuously develop her adventure story.

3 Preliminary User Study

We conducted a qualitative study with 6 participants to evaluate how *WrightHere* supports creative writing. This included four elementary school students: a nine-year-old female (P1), a 10-year-old male (P2), and two 12-year-olds, one male (P3) and one female (P4). Also, one 14-year-old female middle school student (P5) and one 18-year-old female high school student (P6) participated in the study. Each participant took part in a one-hour story creation session. Subsequently, an exploratory interview was conducted with each participant to gather subjective feedback. Participants were asked a few open-ended questions, such as their positive and negative experiences during the study. All sessions were conducted in Korean with parental presence for minor participants. The Institutional Review Board (IRB) of the researcher’s affiliated institution granted approval for the study.

Our user study revealed highly positive responses to *WrightHere* across all participants. The system successfully supported creative writing by helping participants externalize their imagination and develop richer narratives. Participants expressed strong engagement with the system. P2 noted their continued motivation to develop stories: “It is my story, but I want to know what happens next. It’s fun.” P3 also noted how the system’s understanding seemed to deepen as the story progressed, providing “the exact inspiration needed at a specific point.” P5 particularly highlighted the system’s role in creative support, describing it as “feeling like pulling out the

Table 1: Participant Information

Participant	Age	Gender	School Level
P1	9	Female	Elementary
P2	10	Male	Elementary
P3	12	Male	Elementary
P4	12	Female	Elementary
P5	14	Female	Middle School
P6	18	Female	High School

imagination from inside my head.” We examine how participants used *WrightHere* through detailed analysis of participant quotes and behavioral observations. In this section, we discuss the benefits and possible improvements of the system, typically focusing on how participants utilized the generative AI captibility of it.

3.1 Exploring dynamic 3D environments facilitates narrative development

Generating *Story Scenes* and entering into them were the most frequently utilized features, with participants naturally progressing their narratives through spatial exploration. With the system’s ability to quickly generate and switch between scenes, participants would start with simple premises (“went camping”), then develop them further (“went camping with friends,” “went camping with friends when a monster appeared”). This easy scene generation and navigation capability encouraged experimentation and helped maintain creative flow. Notably, while initial writing still showed careful iteration, participants were able to establish a narrative flow quickly, increasing their writing pace. This reduced initial barrier was shown most significantly with P2 and P3, who responded that they not only improved in writing volume but also showed much more confidence in storytelling and writing. When the environment visualization was too random, participants often abandoned

the scene and tried to generate a new one that better aligned with their internal imagination by describing the background more in detail. However, when the visualization was aligned with their internal thoughts, they expressed that it enhanced creativity by transforming abstract thoughts into concrete visuals. For instance, P6 demonstrated how generated environments enriched basic concepts: "Even if I was simply thinking of 'forest,' seeing vivid elements such as the tropical trees, flowers, and lake helped expand my imagination and continue the story." Regarding background music, participants found the preset library of nine tracks too limited for their creative needs. Users specifically wanted more control over their story's audio atmosphere, suggesting integration with external music platforms. This feedback indicated that diversified audio environments also play a significant role in user perception when exploring 3D *Story Scenes*.

3.2 Character creation established foundations for interactive narrative engagement

Although character generation involved noticeable latency, participants valued how these materialized characters served as anchors for dialogue and interaction. The physical presence of characters in the 3D environment provided users with concrete entities to engage with, naturally leading to deeper narrative development through conversation and interaction. Managing latency emerged as a critical consideration in generative AI systems - the faster scene generation was used more frequently than character creation, highlighting how processing speed could directly impact feature usage patterns. Despite the latency, characters proved highly valuable once created due to their dynamic responsiveness to user input, leading to extended conversation sessions. Participants requested more control over character appearance and positioning, particularly noting size discrepancies between characters and environments. While our low-polygon visual style aimed to ensure appropriateness for children, showing alternative character styles during interviews revealed varying preferences among users, suggesting the need for customization options to accommodate diverse aesthetic preferences in future iterations.

3.3 Character interactions catalyze both emotional resonance and story progression

Emotional engagement and story development emerged naturally through character conversation. P1 created a new 'girl' character as a companion after the rabbit character expressed loneliness, while P3 spontaneously contemplated a monster's sad history upon encountering it in a vacant space. For story progression, P2 advanced his narrative by asking their magical sword "Can I fly with you?". While these interactions proved powerful, P3 noted room for improvement, suggesting that character responses should better match their roles: "Even evil monsters speak too nicely; their dialogue should match their character." We also incorporated text-to-speech (TTS) AI to enhance interaction, allowing characters to respond in both text and audio using a friendly voice. The text-to-speech audio feature for character dialogue proved particularly effective, with participants showing increased engagement when characters responded with voiced dialogue. Participants appreciated the addition of audio but suggested improvements to make

the voice more natural. While the diversity of tones was limited, most users did not express a strong need for greater tonal variation. All participants had prior experience with LLMs, and thus they initially approached the system with some skepticism. However, when the system generated one or two responses that they genuinely liked, there was a clear turning point. Still, how participants interact with generative AI was also significantly different - younger participants used it with more unconstrained imagination, while older participants tended to expect and hope for certain output from generative AI. For younger participants, even when the character's response differed slightly from their expectations, participants began to perceive the character naturally as a unique individual that has different thought. They treated the character as a separate being, asking questions like, "How are you feeling?" or "What do you like?"

3.4 Story materials bridged ideation and writing while maintaining creative agency

The system effectively helped translate imagination into written content. P6 highlighted how the memo system helped overcome writer's block: "When stuck, using saved notes helped continue the story instead of wasting too much time." P2 actively used the dialogue materials for co-writing, keeping useful segments while discarding others. Using *Story Materials* during writing and utilizing the co-writing features enhanced the translation of imagination into written expression while maintaining creative control. The usage of this function among participants was particularly insightful. Initially, many users focused on creating characters, dialogues, and scenes without utilizing the memo function. However, once they discovered its writing capabilities, they began using it more actively. P1 started leaving memos whenever she had ideas, questions, or confusions, using them to later refine her story. P2, after seeing the satisfying results of the generated content, revisited all his scenes to leave memos, which he later expanded using the system. P3 expressed high satisfaction with the system's ability to generate content closely aligned with his thoughts, finding it exciting and highly motivating. These behaviors highlight the system's effectiveness in bridging the gap between planning and writing, empowering users to transform their ideas into engaging stories.

4 Limitations and Future Work

Our study has several limitations that suggest directions for future research. First, while our qualitative findings are promising, future work should incorporate quantitative measures [5, 32] examining key metrics such as writing performance, confidence levels, and ideation capacity. This would enable more robust comparison across different user demographics and provide clearer evidence of the system's impact. Second, although our implementation demonstrates the potential of 3D environments for creative writing, the current PC-based interface limits full spatial interaction. While we developed a VR HMD version, it had usability challenges for children with current VR hardware. We therefore focused on validating our core creativity support concepts through a more accessible PC interface, leaving fuller immersive experiences [15] for the future when child-friendly VR hardware becomes available. Several participants expressed interest in this direction, with P2 noting

“I would definitely use this in VR” and P4 expressing excitement about “experiencing the spaces I created.” These responses suggest promising opportunities for spatial computing applications in creative writing, particularly in leveraging the immersive presence that VR environments [34, 37] can uniquely provide. Third, while our current study focused on Korean language users and relatively short one-hour sessions, future work should examine the system’s effectiveness across different linguistic and cultural contexts, and through longer-term engagement.

5 Conclusion

Our study of *WrightHere* demonstrates how AI-infused interactive environments can support creative writing processes. The system’s dual-mode design effectively supported both planning and translating stages of writing, with dynamic 3D environments facilitating narrative development and character interactions driving emotional engagement. The ability to seamlessly transition between writing and exploration helped users maintain creative flow while developing their stories. Key findings revealed that scene exploration and character interaction served as powerful catalysts for story development. While technical challenges like generation latency impacted feature usage patterns, participants found significant value in the system’s ability to transform abstract ideas into concrete visual and interactive elements. The story materials system proved particularly effective in bridging the gap between ideation and writing, helping users overcome creative blocks while maintaining their agency in the writing process. Our observations also highlighted areas for future development, including the need for optimized generation speeds, more diverse audio options, and customizable character aesthetics. While our current implementation focused on PC-based interfaces, participant feedback suggests promising opportunities for future iterations in more immersive platforms as child-friendly VR hardware becomes available. Future research should also examine the system’s effectiveness across different linguistic and cultural contexts through longer-term engagement and quantitative evaluation methods.

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